Seceon EDR / CCE System Health & Reliability Audit

Operational Health, Customer Impact Analysis & Remediation Plan

Date: September 16, 2026
Target Host: 192.168.9.93
Status: Action Required

OPERATING SYSTEM	HARDWARE ARCHITECTURE	APPLICATION STACK	SERVER UPTIME
Rocky Linux 9.5 (5.14.0)	Core i9-14900K (24C/32T), 80GB RAM	Seceon EDR / CCE v12.3.3 (14 Containers)	19 Days, 17 Hours

Executive Summary

A comprehensive live inspection of host **192.168.9.93** running the **Seceon EDR / CCE v12.3.3** stack was conducted. While core container services remain operational, **several critical issues were identified that directly impede customer security operations, cause silent endpoint telemetry loss, and produce onboarding failures**.

Identified Issue	Component	Severity	Direct Customer / Business Impact
1. Remediation Daemon Crashed	`cce-remediation`	CRITICAL	All endpoint remediation actions (host isolation, process termination, quarantine) fail immediately.
2. OpenSearch Field Type Mismatch	`sedr-endpoint-api`	CRITICAL	Silent telemetry data loss for tenant `BHG7200`. Drive observability documents rejected with HTTP 400.
3. Missing Tenant Enrollment Indices	`sedr-endpoint-api`	CRITICAL	Tenants vys3798, pub5493, and agx6974 receive HTTP 404 during agent enrollment and check-in.
4. Monitor Crash & Stack Smashing	`cce-system-health`	HIGH	FIM policy updates fail; server-level health alerting daemon crashes with stack smashing.
5. Kong Gateway SSL SNI Storm	`sedr-kong-1`	MEDIUM	Endless SSL errors on port 8443 as agents use IP literals in SNI. High log I/O overhead and latency risk.
6. OpenSearch Degraded to "Yellow"	`sedr-opensearch-1`	MEDIUM	11 unassigned replica shards on a single-node cluster; index lifecycle & rollup stalls.
7. 31.7 GB Unused Build Cache	Host Docker Engine	MEDIUM	Over 31 GB of orphaned cache layers consuming NVMe drive headroom.

Detailed Findings & Root Cause Analysis

CRITICAL **1. EDR Direct Remediation Daemon Crashed on Startup** Container: `cce-remediation`

Root Cause: The container has conflicting Python packages installed: legacy `websocket (0.2.1)` and `websocket-client (0.57.0)`. When `/usr/local/edr_rem/sdk/polylogyx_apis/api.py` calls `from websocket import create_connection`, Python imports the legacy package which lacks the method, crashing the daemon.

```
Traceback (most recent call last):
  File "/usr/local/seceon/wrapper_edr_direct_remediation.py", line 4, in <module>
    import edr_direct_rem
  File "/usr/local/edr_rem/sdk/polylogyx_apis/api.py", line 19, in <module>
    from websocket import create_connection
ImportError: cannot import name 'create_connection' from 'websocket'
```

Customer Impact: Complete failure of active threat containment. When an analyst triggers an emergency action (e.g. isolating an infected workstation, terminating malicious processes, or quarantining ransomware), the remediation command is never dispatched. The customer network remains exposed.

Remediation: Remove the conflicting dummy package and restart the container:

```
docker exec -it cce-remediation pip uninstall -y websocket
docker restart cce-remediation
```

CRITICAL

2. Telemetry Ingestion Dropped by OpenSearch Mapping Conflict

Container: `sedr-endpoint-api` / `sedr-opensearch-1`

Root Cause: In index `seceon_log_result_bhg7200` (tenant `BHG7200`), field `details.FileSystemMonitoring.Drives.FreeGB` was statically indexed as a `long` integer. When agent `DESKTOP-1Q8NT5M` reports decimal gigabytes (e.g. `468.17`), OpenSearch rejects the document with HTTP 400.

```
Error while inserting the data: ResponseError: status: 400
illegal_argument_exception: mapper [details.FileSystemMonitoring.Drives.FreeGB] cannot be changed from
type [long] to [float]
Path: /seceon_log_result_bhg7200/_doc/ef4cfa75-1b39-4f99-9bac-453dbda128f8
```

Customer Impact: Silent, ongoing observability data loss for tenant `BHG7200`. Endpoint telemetry and disk health events fail to record. Customer dashboards display zero or stale metrics, and low disk space warnings will not trigger.

Remediation: Update the index template so `FreeGB` is mapped as `float` or `double`:

```
curl -X PUT "http://127.0.0.1:9200/_template/seceon_log_result_template" \
  -H 'Content-Type: application/json' \
  -d '{"index_patterns":["seceon_log_result_*"],"mappings":{"properties":{"details":{"propertie
s":{"FileSystemMonitoring":{"properties":{"Drives":{"properties":{"FreeGB":{"type":"float"},"Tota
lGB":{"type":"float"}}}}}}}}}'
```

CRITICAL

3. Missing Enrollment Indices Causing HTTP 404 Agent Check-in Failures

Container: `sedr-endpoint-api`

Root Cause: The API repeatedly throws 404 errors when endpoints under tenants `vys3798`, `pub5493`, and `agx6974` attempt to check in, because their corresponding enrollment indices do not exist in OpenSearch.

```
Error: ResponseError: index_not_found_exception: no such index [seceon_endpoint_enroll_vys3798]
Error: ResponseError: index_not_found_exception: no such index [seceon_endpoint_enroll_pub5493]
Error: ResponseError: index_not_found_exception: no such index [seceon_endpoint_enroll_agx6974]
```

Customer Impact: Agents for these three customer organizations cannot enroll, renew security tokens, or validate policies. Security teams believe protection is deployed, but endpoints are disconnected from the platform.

Remediation: Re-trigger tenant provisioning from the Seceon management console or recreate the missing enrollment indices in OpenSearch.

HIGH

4. System Health Monitor & FIM Sync Process Crash

Container: `cce-system-health-monitor`

Root Cause: The startup script fails to locate `/copyOldFimConfig.py`, followed by an immediate stack corruption termination in an auxiliary monitoring binary.

```
python3: can't open file '/copyOldFimConfig.py': [Errno 2] No such file or directory
*** stack smashing detected ***: terminated
```

Customer Impact: FIM (File Integrity Monitoring) configuration synchronization fails. System-level resource alerts (e.g. host memory exhaustion, disk full triggers) fail to report to tenant dashboards.

Remediation: Correct the missing script path in container startup scripts and audit compiled helper tools in `/usr/local/addondevices/` for memory safety.

MEDIUM

5. Kong Gateway Flooded with SSL SNI Errors from Endpoint Agents

Container: `sedr-kong-1`

Root Cause: Agents connecting from `192.168.19.x` populate the TLS Server Name Indication (SNI) extension with the server's IP address (`192.168.9.93`). RFC 6066 mandates that SNI must be a hostname; Kong's Lua SSL filter logs an error on every request.

```
[error] [lua] certificate.lua:33: log(): [ssl] failed to fetch SNI: failed to fetch '192.168.9.93' SN
I: [off] must not be an IP, client: 192.168.19.85, server: 0.0.0.0:8443
```

Customer Impact: Millions of log entries waste server disk I/O and CPU time. Additionally, strict client TLS configurations may reject connections when SNI negotiation fails.

Remediation: Assign a DNS FQDN to the EDR server (e.g. `edr.internal.net`) and configure the agent installer/profile to point to the FQDN rather than the raw IP.

MEDIUM

6. OpenSearch Cluster Degraded to "Yellow" (11 Unassigned Shards)

Container: `sedr-opensearch-1`

Root Cause: Several indices were initialized with `number_of_replicas: 1` on a single-node cluster. Replica shards cannot be allocated to the same node as their primary, causing permanent cluster degradation.

Customer Impact: Cluster health is perpetually flagged as degraded. Scheduled ISM (Index State Management) rollups and purge policies may stall.

Remediation: Set replica count to 0 for all indices on this standalone node:

```
curl -X PUT "http://127.0.0.1:9200/*/_settings" \
  -H "Content-Type: application/json" \
  -d '{"index": {"number_of_replicas": 0}}'
```

MEDIUM

7. 31.7 GB Reclaimable Docker Build Cache on Host

Host Storage: `/var/lib/docker`

Root Cause: Over 1,099 obsolete build cache layers occupy **31.68 GB** of NVMe storage from previous container builds.

Customer Impact: Reduced disk headroom increases the risk of root partition exhaustion as OpenSearch indices expand.

Remediation: Reclaim disk space by running:

```
docker builder prune -a -f
```

Immediate Action Roadmap

Phase	Action Item	Estimated Effort	Expected Result
Phase 1 (Immediate)	Uninstall `websocket` package inside `cce-remediation`	2 minutes	Restores active endpoint threat remediation and containment.
Phase 1 (Immediate)	Execute replica count reset on OpenSearch (`number_of_replicas: 0`)	1 minute	Turns OpenSearch cluster status to **GREEN**.

Phase	Action Item	Estimated Effort	Expected Result
Phase 2 (Day 1)	Update index template for `seceon_log_result_*` (FreeGB float)	15 minutes	Stops endpoint telemetry dropping for tenant `BHG7200`.
Phase 2 (Day 1)	Re-provision enrollment indices for `vys3798`, `pub5493`, `agx6974`	30 minutes	Restores enrollment and heartbeats for affected tenants.
Phase 3 (Day 2)	Deploy DNS FQDN for EDR server and update agent configurations	1 hour	Eliminates millions of SNI errors in Kong API Gateway.
Phase 3 (Day 2)	Prune 31.7 GB Docker build cache	2 minutes	Reclaims 31.7 GB NVMe disk space.
